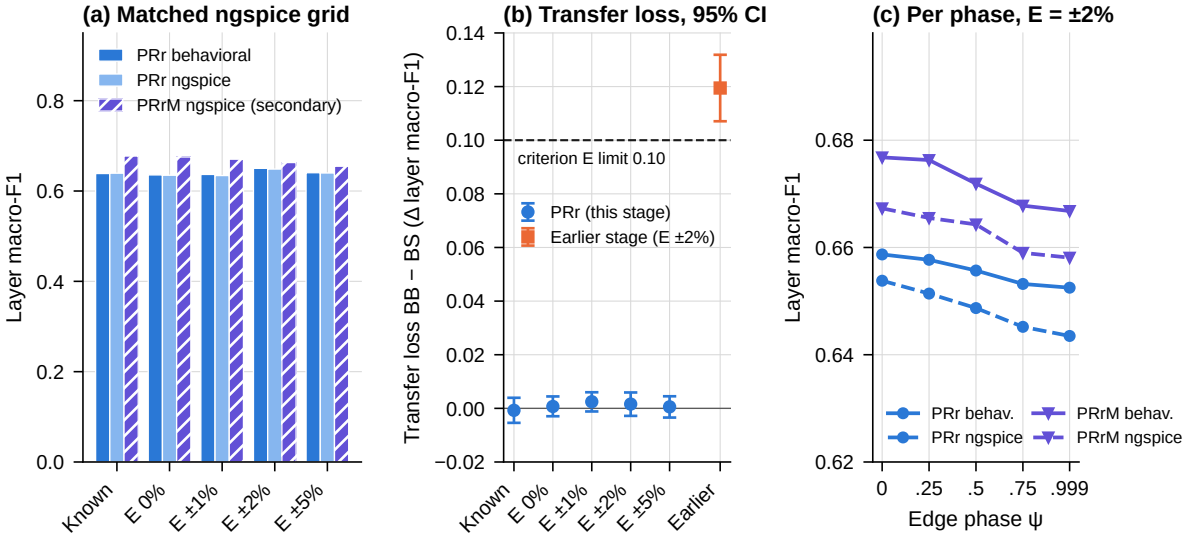
(a) Matched ngspice grid
PRr behavioral
PRr ngspice
PRrM ngspice (secondary)
Layer macro-F1
0.0
0.2
0.4
0.6
0.8
Known
E 0%
E ±1%
E ±2%
E ±5%
(b) Transfer loss, 95% CI
Transfer loss BB − BS (Δ layer macro-F1)
criterion E limit 0.10
PRr (this stage)
Earlier stage (E ±2%)
−0.02
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Known
E 0%
E ±1%
E ±2%
E ±5%
Earlier
(c) Per phase, E = ±2%
Layer macro-F1
0.62
0.64
0.66
0.68
PRr behav.
PRr ngspice
PRrM behav.
PRrM ngspice
0
.25
.5
.75
.999
Edge phase ψ